POSE EMBED

Understanding and defending the experiment

A study guide from movement data to a fair scientific claim

Theory, design decisions, code provenance, and verification

The question

Does the full contextual training recipe lose less one-shot retrieval accuracy than standard contrastive training when the query's pose information is deliberately damaged?

This guide assumes comfortable programming, matrix algebra, multivariable calculus, and introductory probability and statistics. It develops the machine-learning ideas needed for this study rather than reviewing basic coding.

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How to study this guide

The goal is to explain *what is being measured, why the comparison is fair, and what evidence permits a conclusion.* You do not need to memorize test names, cluster job numbers, hashes, or every historical implementation detail.

Read Sections 1–8 for the scientific account. Section 9 explains why working software is not enough and what has actually passed. Section 11 prepares you for questions. The appendices contain short answers, a notation reference, and source links.

At each equation, identify the objects, their shapes, and what changes during training. At each experimental decision, ask: *what alternative explanation does this control remove, and what uncertainty remains?*

The three companion notebooks use invented, small arrays. They illustrate geometry and gradients, fair comparisons, and paired cluster resampling. They are learning exercises, not an alternate research pipeline or evidence that the real study has passed a gate.

Three different kinds of statement

Design: what the locked study requires. **Evidence:** what a dated run and its verification established. **Interpretation:** what may reasonably be concluded from that evidence.
For example, three complete engineering pilots are evidence that those runs worked under their recorded conditions. They are not the planned final comparison, and they do not automatically validate a changed runtime.

The guide uses the current frozen v3 study. The earlier 26-method, fine-tuned v2 benchmark is preserved history, not completed work or a requirement of this December comparison. The October 28 operational checkpoint is on a preparation branch; its local launcher changes are not claimed to have run on the cluster.

1 The question and the unit of evidence

1.1 Start with a concrete task

Suppose a gallery contains one clean motion example for each of twenty actions. A query is another motion clip. The system ranks the gallery by similarity and returns the action of the closest example. A sitting-down query should retrieve the sitting-down example even when some joint observations are unreliable.

This is **retrieval**: find the matching example. It is not a system that estimates joint positions from raw video, predicts rowing force, or judges whether an action was performed well. The input already contains estimated joint positions. The action label provides the definition of a correct match; it does not measure every possible meaning of biomechanical similarity.

One-shot describes the gallery: one labeled example for each new action. It does not mean the model learns from one training clip. The comparison rule is learned using thousands of clips from other actions, then applied to actions excluded from this study's head training and model selection.

Write a motion example as x, its action label as y, and its learned representation as $z = f(x)$. Let $\mathcal{A}$ be the twenty gallery action labels and g_a the reference clip for action a. Then

$$\widehat{y}(x) = \arg\max_{a \in \mathcal{A}} s(f(x), f(g_a)).$$

The function s is cosine similarity. Later sections derive both f and s. No final-test labels are used to update f.

1.2 The hypothesis is deliberately narrower than "better AI"

There are three training recipes: standard contrastive, full contextual, and supervised contrastive (SupCon). The one confirmatory comparison is contextual versus standard contrastive. SupCon provides a descriptive reference.

Each trained head is evaluated on clean queries and on nine damaged-query conditions. For a method m, seed s, and corruption condition c, define

$$d_{m,s,c} = A_{m,s,0} - A_{m,s,c},$$

where A is top-1 accuracy and 0 denotes clean input. The main effect is

$$\Delta = \frac{1}{3 \cdot 9} \sum_{s,c} (d_{\text{contextual},s,c} - d_{\text{contrastive},s,c}).$$

Negative Δ means the contextual recipe loses fewer accuracy points under the declared damage mixture. A later section explains uncertainty and the prespecified decision rule.

A smaller drop is not necessarily better damaged-input accuracy

Invented example: standard contrastive goes from 90% clean to 75% damaged, a 15-point drop. Contextual goes from 80% to 70%, a 10-point drop. Then $\Delta = 10 - 15 = -5$ percentage points. Contextual degrades less, but its damaged-input accuracy is five points worse. Both statements must be reported.

The hypothesis concerns a *full recipe*. Its contextual term, positive margin, and similarity regularizer differ from the standard recipe. A positive finding cannot identify the neighborhood term alone as the

cause. The omitted extra control limits causal attribution; it does not make the declared recipe-to-recipe comparison meaningless.

1.3 Why use this study design?

Freezing an existing motion encoder makes the training comparison narrower and less expensive than fine-tuning the whole network. Holding the head architecture, data, duration, and tuning budget fixed makes differences easier to attribute to the training recipes. Testing unfamiliar action classes asks whether the learned comparison transfers beyond the action labels used for training.

Damaging only queries keeps the references unchanged. The experiment can then measure sensitivity to the query disturbance. It does not establish performance when both gallery and query are damaged, or robustness to arbitrary camera failures. Those would be different experiments.

Source trail: Final v3 plan, Sections 1, 3–6; Decision 0004.

2 Data: what one row actually means

2.1 Dataset, observations, and dependence

The dataset family is NTU RGB+D 120,[1] but this pipeline uses its adopted **HRNet-derived 2D pose annotations**. The model channels are x coordinate, y coordinate, and confidence. They are not x, y, z ground-truth 3D positions. "Clean" means before this study adds damage; the pose estimates already have their own errors.

The adopted inventory has 113,945 usable captures. Together with 535 documented missing-skeleton exclusions, these reconcile to 114,480 nominal captures. Reconciling the count matters because a changed input inventory could silently change the population being compared.

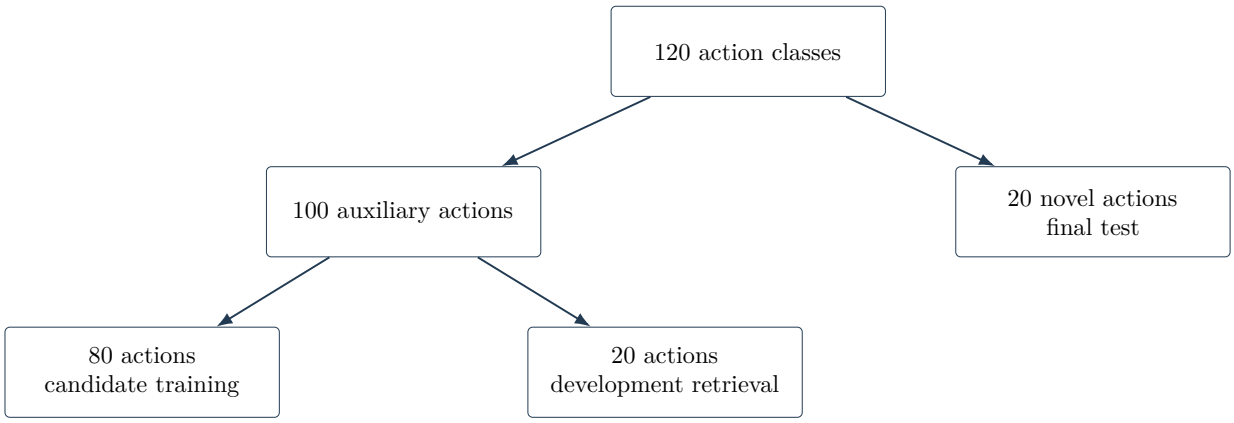
A capture has metadata including setup, performer, repetition, action, and camera. Several cameras can observe the *same underlying performance*. Thus three rows need not be three independent examples. The performance-group key is

$$(\text{setup}, \text{performer}, \text{repetition}, \text{action}),$$

with camera deliberately omitted. This key matters twice: when excluding near-duplicate gallery performances from primary queries, and when estimating uncertainty.

2.2 Split by action before learning the comparison

The 120 action classes are partitioned into 100 auxiliary actions and twenty novel actions. Within the auxiliary set, eighty actions train candidate heads and twenty different actions support development evaluation.



Role	Rows	Permitted use
Candidate-head training	76,013	Learn weights using eighty actions.
Development-validation pool	18,988	Twenty held-out auxiliary actions.
Development gallery	20	One fixed reference per development action.
Primary development queries	18,929	Evaluate candidates without gallery-performance views.
Final-head training	95,001	All 100 auxiliary actions, after selecting recipes.
Novel gallery	20	Fixed official exemplars.
Primary novel queries	18,884	Confirmatory final evaluation.
Exact-official novel queries	18,924	Descriptive compatibility evaluation.

These rows are *roles*, not disjoint quantities to add together. Final-head training is the union of the two auxiliary pools: $76{,}013 + 18{,}988 = 95{,}001$. Gallery and query roles are selected from their respective validation or novel pools. The twenty novel anchors plus 18,924 official queries give 18,944 novel rows; $95{,}001 + 18{,}944 = 113{,}945$.

After choosing one learning rate for each recipe, nine fresh final heads are trained on all 100 auxiliary actions. Reusing the development actions at this point is legitimate because recipe selection is finished and the novel actions still provide the final evaluation. The final heads are not the candidate or pilot checkpoints with new names.

2.3 Why two query definitions?

The exact-official query set removes the gallery exemplar itself. The primary set also removes other camera views of that exemplar's performance. Otherwise, the system could retrieve a simultaneously recorded version of the same event, which is easier than recognizing another performance of the action.

The two definitions answer related but different questions. The official one maintains benchmark comparability; the primary one reduces this specific cross-camera dependence. Their difference is not selected after seeing scores. Exact IDs are locked from metadata. Do not infer counts by assuming every performance has three valid camera views.

This is an **action-disjoint** study, not a claim of performer-disjoint generalization. People or setups can recur. "Novel" describes the role in this study's head training and selection; it does not prove the pretrained encoder has never encountered any related action in any source data.

2.4 A manifest is part of the experimental definition

A manifest records which sample IDs belong to a role, their order, exclusions, and source bindings. Its hash is a compact fingerprint of those exact bytes. Recreating a manifest independently and comparing it checks that the stated rules lead to the same population.

This is stronger than checking a row count. Two files can each contain 18,884 rows but describe different queries. Conversely, equal hashes establish equality of content, not that the content is scientifically correct. Both semantic validation and identity checks are needed.

Check your understanding

Why would a random split of capture rows fail to test the intended transfer to new actions? Why is removing only the exact gallery filename insufficient? Answers: Questions 1–2 in the answer key.

Source trail: Final v3 plan, Section 3; Adopted manifest audit; Input contract amendment.

3 From joint observations to a comparison vector

3.1 Follow the shapes

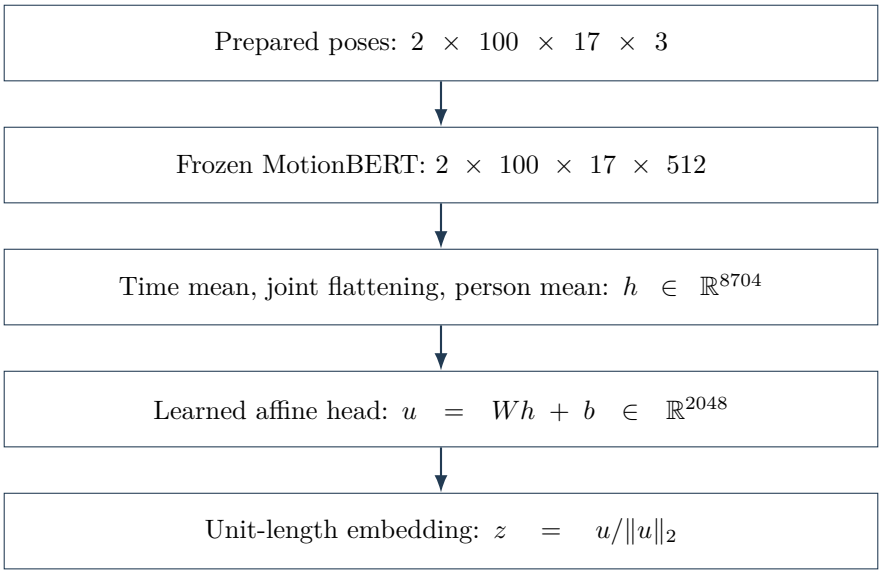
One deterministically prepared example has shape

$$X \in \mathbb{R}^{2\times100\times17\times3}.$$

The axes are person slot, sampled frame, joint, and $(x, y, \text{confidence})$. A missing second person is represented by a padded slot. Temporal sampling selects 100 frame positions; it is not a claim that the original recording lasted 100 frames or that all clips had the same physical duration.

The preparation sequence is camera normalization, person tracking, conversion from the COCO joint convention to the H36M joint convention, uniform temporal sampling, padding, and spatial normalization. Tracking tries to keep person slots consistent over time. Conversion produces the joint arrangement expected by the pretrained model. Normalization removes some dependence on image position and scale; it does not eliminate viewpoint, tracking, or pose-estimation errors.

Some converted joints combine several source joints. Their coordinates can be averaged, but the local implementation uses the *minimum* confidence of the contributing joints. If one necessary source is absent, its derived joint should not acquire high confidence merely because another source is visible. This is an explicit departure from upstream confidence handling.



3.2 What MotionBERT contributes

A representation is a numerical description useful for a later task. Raw coordinates describe where joints were observed. Learned features can combine positions across joints and time so that the next stage need not learn all motion structure from scratch.

MotionBERT was pretrained to recover motion structure from partial or noisy 2D observations using a 3D reconstruction objective. This study reuses its generic pretrained encoder. The adapter requests its internal representation and bypasses its 3D output head. There is no newly estimated 3D skeleton that must be evaluated between our input poses and retrieval vectors.[2]

The essential transformer operation is attention. Put a set of token features in a matrix $H \in \mathbb{R}^{L\times d}$, one token per row. Learned matrices produce

$$Q = HW_Q, \qquad K = HW_K, \qquad V = HW_V.$$

The query–key dot product measures which tokens should exchange information. For each row, softmax turns scores into positive weights that sum to one:

$$a_{ij} = \frac{\exp(q_i^\top k_j/\sqrt{d_k})}{\sum_\ell \exp(q_i^\top k_\ell/\sqrt{d_k})}, \qquad o_i = \sum_j a_{ij} v_j.$$

The square-root factor moderates score scale as the key dimension grows. Unlike a fixed temporal average, these weights depend on the input. Multiple attention heads provide several learned ways to mix information, while position information distinguishes joints and time indices.

MotionBERT's spatial attention exchanges information among joints in a frame; temporal attention exchanges information across frames. Its two streams apply spatial and temporal operations in opposite orders and fuse their outputs. You need to understand this role, not reproduce every transformer layer to explain this experiment. The pretrained implementation is imported intact.

Frozen and **evaluation mode** are different controls. Frozen parameters receive no training updates. Evaluation mode selects inference behavior for modules with training-dependent behavior, such as dropout. The project uses both, so all recipes share a fixed feature extractor.

3.3 Why averaging features does not discard all motion information

Let $F_{p,t,j,:}(X) \in \mathbb{R}^{512}$ denote an encoder output. The pre-head vector is

$$h(X) = \frac{1}{2}\sum_{p=1}^{2} \mathrm{vec}_j\left(\frac{1}{100}\sum_{t=1}^{100} F_{p,t,j,:}(X)\right) \in \mathbb{R}^{17\cdot 512}.$$

Here vec_j concatenates the seventeen joint feature vectors. The mean is taken after the encoder has combined information across time. It is therefore not equivalent to averaging raw poses before any motion processing. It still imposes an information bottleneck: whatever the pooled representation does not retain cannot be recovered by the head.

With a frozen encoder, $h(X)$ can be calculated once and stored in a feature cache. This saves repeated encoder work across head-training runs. A cache is valid only for its exact source data, preprocessing, encoder weights, code, and relevant runtime. A damaged query needs a fresh encoder pass on the damaged tensor; adding noise to a cached vector would be a different intervention.

3.4 The trainable head and the geometry of retrieval

The trained transformation is

$$u = Wh + b, \qquad W \in \mathbb{R}^{2048\times 8704}, \quad b \in \mathbb{R}^{2048}, \qquad z = \frac{u}{\|u\|_2}.$$

There are $2048(8704 + 1) = 17{,}827{,}840$ trainable parameters. The architecture is simple, but its parameter count is not tiny. The cached h vectors are *not* unit-normalized; the normalization follows the learned projection. Zero projections are an invalid numerical state to detect rather than meaningful directions for retrieval.

For nonzero vectors u, v, cosine similarity is

$$\cos(u, v) = \frac{u^\top v}{\|u\|\|v\|}.$$

It compares direction rather than length. Once both vectors have unit norm, $s_{ij} = z_i^\top z_j$. Expanding a squared distance gives

$$\|z_i - z_j\|^2 = z_i^\top z_i + z_j^\top z_j - 2z_i^\top z_j = 2 - 2s_{ij}.$$

Thus sorting by largest cosine is the same as sorting by smallest squared Euclidean distance on these unit vectors. This equivalence is why the contextual neighborhood can be expressed using $\max(2 - 2s_{ij}, 0)$. The maximum only guards against a tiny negative distance from floating-point arithmetic.

A two-coordinate retrieval example

Let a query be $q = (0.8, 0.6)$ and the two gallery vectors be $g_1 = (1, 0)$ and $g_2 = (0, 1)$. All are unit vectors. Their cosine scores are 0.8 and 0.6, so g_1 ranks first. The squared distances are 0.4 and 0.8, producing the same ranking. Multiplying the unnormalized query by ten changes neither its normalized direction nor the ranking.

The output has 2,048 coordinates, not 120 class probabilities. Labels influence the geometry during learning; an unfamiliar action can still be recognized by comparison with its gallery exemplar. Contextual neighborhoods affect training, but **all three recipes use the same cosine retrieval at evaluation**. There is no contextual reranking added at query time.

3.5 From a loss to a weight update

A loss $L(\theta)$ assigns a penalty to the current behavior, with $\theta = (W, b)$. Minimizing it requires a direction in parameter space. For a small perturbation δ,

$$L(\theta + \delta) \approx L(\theta) + \nabla L(\theta)^\top \delta.$$

Taking $\delta = -\eta \nabla L$ lowers this local approximation when $\eta > 0$. That is the basic gradient-descent idea. It does not promise global improvement for an arbitrarily large step, or a global optimum for this nonconvex objective.

Backpropagation is the chain rule applied through the computation. For $z = u/\|u\|$ away from zero,

$$\frac{\partial z}{\partial u} = \frac{I - zz^\top}{\|u\|}.$$

The factor $I - zz^\top$ removes the component parallel to z: changing only magnitude does not change direction. If $g_u = \partial L/\partial u$, then

$$\frac{\partial L}{\partial W} = g_u h^\top, \qquad \frac{\partial L}{\partial b} = g_u.$$

Batch gradients add the contributions prescribed by the loss reduction. The encoder weights do not enter the update. Notebook 1 works through a small normalized-head gradient numerically; its simple loss is explicitly not the project's full training objective. This is ordinary backpropagation through differentiable operations. The hard contextual neighborhood instead uses a prescribed surrogate gradient, explained in the next section.

3.6 Why AdamW, and what is fixed?

The actual optimizer is AdamW. For gradient g_t, it keeps exponentially weighted first and second moments,

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \qquad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2,$$

and corrects their startup bias: $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$. The essential elementwise update is

$$\theta_t = (1-\eta\lambda_w)\theta_{t-1} - \eta \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \varepsilon_o}.$$

The first term is decoupled weight decay; the second uses recent gradient direction and scale. Weight decay is distinct from the contextual recipe's mean-similarity regularizer, which acts on embeddings through the loss.

All recipes use $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon_o = 10^{-8}$, and $\lambda_w = 10^{-4}$ on weight and bias. Learning rate is constant within a run. There is no scheduler, dropout, extra augmentation, gradient clipping, early stopping, or mixed precision. These are fixed controls, not assertions that these choices are universally optimal.

Check your understanding

Explain why a frozen encoder still permits learning, why the head can recognize an action absent from its training labels, and why perturbing cached features would change the experiment. Answers: Questions 3–4.

Source trail: Deterministic preprocessing; Encoder adapter; Pooling and head; Training runner; Pinned DSTformer source.

4 What is reused, adapted, and written for this study

The research contribution is not a new pose detector or a new transformer backbone. It is the specified transfer comparison and the controlled process needed to interpret its result.

Component	Provenance	Role and boundary
Pose data	Existing NTU dataset and HRNet-derived pose annotations distributed through OpenMMLab/PYSKL	Reuse the available estimates. No new HRNet training, video collection, or native 25-joint 3D input.
MotionBERT encoder	Unchanged code imported from a pinned Apache-2.0 checkout; verified pretrained checkpoint	Reuse the architecture and learned weights; neither source clone nor checkpoint is committed into this repository.
Preprocessing	Minimal licensed port of MotionBERT action-data utilities	Make sampling deterministic, preserve confidence through tracking/conversion, and apply the declared confidence correction.

Component	Provenance	Role and boundary
Pooling and head	Ported MotionBERT action-head semantics	Reuse time/person pooling and projection structure; train fresh head weights for our objectives.
Contextual loss	Independently implemented from published equations, developed from the project's teaching implementation	The authors' source repository is reference-only because no license was discovered. It is not copied or adapted.
Contrastive and SupCon	Local implementations of the declared formulas	Make reduction rules, margins, temperature, and single-view semantics explicit.
Optimization engine	PyTorch tensor operations, autograd, and AdamW	Reuse numerical machinery; local code defines which parameters and batches it updates.
Training and evaluation	Local protocol-specific runners	Train on cached auxiliary features; keep novel evaluation out of the training loop.
Experimental controls	Local manifests, sampler, corruptions, locks, provenance, analysis, and report code	Define a reproducible experiment rather than inherit benchmark defaults.

4.1 Why not reuse the upstream one-shot training loop?

That loop evaluates novel classes during training. Our study requires those outcomes to remain unavailable until training, selection, and the final lock are complete. Reusing the encoder is compatible with this requirement; reusing that evaluation schedule is not. The local head-training loop is a necessary change in scientific procedure, not a claim that upstream code is defective.

4.2 Two kinds of provenance

Intellectual provenance identifies whose method, data, and code are being used, along with the permitted reuse and attribution. **Execution provenance** identifies the exact source revision, checkpoint, input IDs, configuration, and runtime that produced an artifact. A result needs both.

Historical v2 contains other methods and upstream sources, but those are not all used in this v3 comparison. The current study does not need a tour of every preserved benchmark implementation. Nor does a source repository being publicly viewable establish permission to copy its code; the project's source manifest and notices record that boundary.

Source trail: Pinned upstream manifest; Third-party notices; Pose-source investigation; Adapter reuse boundary; Contextual source and equation notes.

5 The three training recipes, from first principles

5.1 What a training batch tells the model

For a batch of n normalized embeddings, form the similarity matrix $S = ZZ^\top$, where row i of Z is $z_i^\top$. Let

$$y_{ij} = \mathbf{1}[\text{action}_i = \text{action}_j].$$

A *positive pair* has the same action label; a *negative pair* has different labels. These names describe supervision, not good or bad examples. The losses encourage a geometry consistent with those labels.

Each physical batch has eight action classes and four examples per class: $P = 8$, $K = 4$, $n = PK = 32$. Each anchor has three nonself positives and twenty-eight negatives. The full matrix contains 96 directed positive pairs and 896 directed negative pairs, excluding self-pairs. A directed pair counts (i, j) separately from (j, i); the declared reductions handle that consistently.

Why not draw thirty-two arbitrary examples? An unbalanced random batch may have too few same-label examples to provide the required positive relations. The balanced sampler controls that issue and supplies the local population in which contextual neighborhoods are defined.

5.2 Standard contrastive: enforce similarity margins

We want positive similarities to be sufficiently high and negative similarities to be sufficiently low. Define a hinge $[a]_+ = \max(a, 0)$:

$$h_{ij}^+ = [m_+ - s_{ij}]_+, \qquad h_{ij}^- = [s_{ij} - m_-]_+.$$

For the standard recipe, $m_+ = 0.9$ and $m_- = 0.6$. A positive pair with similarity 0.7 receives a penalty 0.2. A negative pair with similarity 0.7 receives a penalty 0.1. Positive similarity above 0.9 and negative similarity below 0.6 produce no hinge penalty. The interval between the margins is an intentional gap, not a probability threshold for an action classifier.

Define the strictly active sets

$$\mathcal{A}_+ = \{(i, j) : i \neq j,\ y_{ij} = 1,\ h_{ij}^+ > 0\}, \quad \mathcal{A}_- = \{(i, j) : y_{ij} = 0,\ h_{ij}^- > 0\}.$$

Then

$$L_{\text{contrast}} = \frac{\sum_{(i,j)\in\mathcal{A}_+} h_{ij}^+}{|\mathcal{A}_+|} + \frac{\sum_{(i,j)\in\mathcal{A}_-} h_{ij}^-}{|\mathcal{A}_-|},$$

where an empty active set contributes zero.

This detail is consequential: these are separate means over *violations*, not one mean over all pairs. An abundance of already-satisfied negatives does not dilute the remaining negative violations. The hinges are not squared. Away from boundaries and changes in the active set, each active positive gives derivative $-1/|\mathcal{A}_+|$ with respect to its similarity; each active negative gives $+1/|\mathcal{A}_-|$. Through the chain rule these gradients change W, b so the relevant similarities move toward their margins.

5.3 Context: ask how examples relate to the rest of the batch

Direct similarity examines a pair. Contextual similarity also examines its relations to other examples. Two noisy instances can be more convincingly related if their neighborhoods agree. That is a motivation, not a guarantee: bad neighborhoods can also reinforce bad relationships.

The current implementation has five steps: find nearby examples, measure shared neighbors and non-neighbors, keep that agreement local, expand using reciprocal neighbors, then make the score symmetric. The final score is trained to match the same-action indicator y_{ij}. It is not trained to imitate the current cosine matrix; doing that would change the target of the learning problem.

The following equations specify the local implementation. Their source trail is the protocol and independently written local code; the contextual paper provides the method's intellectual origin.[3]

Step 1: define a neighborhood

Compute squared unit-vector distances and a binary membership matrix:

$$d_{ij} = \max(2 - 2s_{ij}, 0), \qquad N_{ij} = \mathbf{1}[d_{ij} \le d_{i,(k)} + \epsilon].$$

$d_{i,(k)}$ is the kth smallest distance in row i. Here $k = 4$ and $\epsilon = 0.05$. **Self is included** in the ordering. The tolerance admits points just beyond the kth distance, so the neighborhood may contain more than four examples, especially with ties.

The symbol k is a neighbor rank used to set the distance cutoff; K is examples per class. Both equal four in the fixed recipe, but their conceptual roles differ. Labels do not choose the neighbors. Current embedding distances do.

Step 2: compare neighborhood membership

For row i, set

$$a_i = \max\left(\sum_\ell N_{i\ell}, 1\right), \qquad b_i = \max\left(\sum_\ell (1 - N_{i\ell}), 1\right).$$

The first-stage score is

$$A_{ij} = \frac{1}{2}\left(\frac{\sum_\ell N_{i\ell}N_{j\ell}}{a_i} + \frac{\sum_\ell (1 - N_{i\ell})(1 - N_{j\ell})}{b_i}\right) N_{ij}.$$

The numerators count shared neighbors and shared non-neighbors. The row denominators turn them into fractions relative to i. Multiplication by N_{ij} restricts the score to pairs already in i's neighborhood. The clamp keeps an empty complement from creating division by zero; it is a declared local finite convention.

Work one neighborhood calculation by hand

In an invented six-example batch, let $i = 1$ have neighbors $\{1, 2, 3, 4\}$ and $j = 2$ have neighbors $\{1, 2, 3, 5\}$. They share three of i's four neighbors. Their non-neighbor sets are $\{5, 6\}$ and $\{4, 6\}$, sharing one of two. Because $N_{12} = 1$,

$$A_{12} = \tfrac{1}{2}(3/4 + 1/2) = 0.625.$$

This is only the first-stage score, not the final contextual similarity or the cosine similarity. With unequal row sizes, A_{ij} need not equal A_{ji}.

Step 3: reciprocal expansion and symmetry

Construct another neighborhood $N^{1/2}$ using the $k/2$ threshold and the same tolerance. A reciprocal relation requires both directions:

$$R_{i\ell} = N^{1/2}_{i\ell} N^{1/2}_{\ell i}.$$

Use those reciprocal neighbors to average first-stage rows, then symmetrize:

$$B_{ij} = \frac{\sum_\ell R_{i\ell} A_{\ell j}}{\max(\sum_\ell R_{i\ell}, 1)}, \qquad C_{ij} = \frac{B_{ij} + B_{ji}}{2}.$$

This incorporates the local relationship profiles of mutually close examples. Symmetry gives a common score for the pair regardless of direction.

Step 4: fit action-label agreement

The contextual penalty is

$$L_{\text{context}} = \frac{1}{n^2} \sum_{i \neq j} (C_{ij} - y_{ij})^2.$$

It asks contextual similarity to approach one for same-action pairs and zero for different-action pairs. Self terms are omitted, but the denominator remains n^2, not $n(n-1)$. Even a small reduction change can alter gradient scale and therefore the relative weight of the objectives.

5.4 How can a hard neighborhood comparison have a gradient?

An indicator is locally constant almost everywhere and discontinuous at its boundary. Its ordinary derivative cannot provide useful smooth updates for moving points between neighborhoods. The recipe deliberately uses a **straight-through estimator** (STE): perform the hard comparison in the forward pass, but prescribe a surrogate derivative in the backward pass.

For the comparison $H(x, y) = \mathbf{1}[x \geq y]$, the implemented rule assigns

$$\frac{\widetilde{\partial} H}{\partial x} = \alpha, \qquad \frac{\widetilde{\partial} H}{\partial y} = -\alpha, \qquad \alpha = 10.$$

The tilde emphasizes that this is a chosen backward rule, not the ordinary derivative of the hard function. In the neighborhood expression, $x = -d_{ij} + \epsilon$ and $y = -d_{i,(k)}$.

The kth-distance thresholds and first-stage counts are *detached*: they retain their forward values but are treated as constants during differentiation. The reciprocal denominator in B is *not* detached. It participates in the backward calculation. These choices are part of the fixed method, not details that an optimizer can infer from a prose description.

What a derivative check can and cannot establish

For a smooth head calculation, finite differences can check an analytic gradient. For this hard comparison, a finite difference will usually see zero while the prescribed STE returns a nonzero value. That disagreement is expected. Verification must check the declared surrogate backward rule independently; it must not claim the rule is the true derivative of the hard forward function.

5.5 The full contextual recipe includes two more terms

The recipe also contains the margin contrastive loss and a mean-similarity regularizer:

$$L_{\text{reg}} = \left(\frac{1}{n^2} \sum_{i,j} s_{ij} - 0.25 \right)^2, \qquad L_{\text{full}} = 0.4L_{\text{context}} + 0.6L_{\text{contrast}} + 0.1L_{\text{reg}}.$$

The regularizer includes diagonal entries. Its target controls average batch similarity; it is not an action probability or a guarantee against collapse. The contrastive term here uses $m_+ = 0.75$ and $m_- = 0.6$, whereas the standalone contrastive arm uses $m_+ = 0.9$.

That means a contextual win could arise from the combined effects of neighborhood structure, a different margin, and regularization. Separating their contributions would require matched ablations, including the omitted control. This study instead compares the two complete declared recipes within its available scope.

The fixed values transfer a published recipe into a pose study. They were not established as optimal pose hyperparameters. Do not substitute defaults from an older teaching class or historical protocol for the explicit v3 configuration.

5.6 SupCon: rank all positive examples against the batch

Let $P(i) = \{p \neq i : y_{ip} = 1\}$. SupCon assigns a softmax distribution over every nonself example and penalizes the log probability of each positive:

$$L_{\text{SupCon}} = -\frac{1}{n} \sum_i \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(s_{ip}/\tau)}{\sum_{a \neq i} \exp(s_{ia}/\tau)}, \qquad \tau = 0.07.$$

For one anchor, every positive should rank well relative to all alternatives. The denominator includes other positives as well as negatives; self is excluded. Using the log converts a product-like probability objective into a sum and strongly penalizes a positive assigned very small probability.

Temperature τ sets sensitivity to score gaps. Two alternatives whose similarities differ by 0.14 have softmax weight ratio $\exp(0.14/0.07) = \exp(2) \approx 7.39$. A smaller temperature makes the same gap matter more. This does not make the value a calibrated probability that an arbitrary final query has a particular action.

The study uses a single-view adaptation: no second augmented view and no extra temperature multiplier. SupCon is supervised because action labels define positives. It is not being used as a label-free pretraining method here. In the ideal limit of zero probability on negatives, equal allocation among three positives gives per-anchor loss $\log 3$ rather than zero. Raw loss values from different recipes are therefore not a common accuracy scale.[4]

5.7 What fairness does and does not require

All recipes receive the same architecture, data, batch plans, initialization within a seed, training duration, and learning-rate grid. They are allowed to use different objective formulas because that is the treatment being compared. Equal tuning opportunities do not mean equal floating-point operation counts, nor that each method reaches its own global optimum.

Recipe	Main training signal	Interpretation boundary
Contrastive	Fix positive/negative margin violations	A specified baseline, not every possible contrastive method.
Full contextual	Neighborhood agreement, contrastive margins, mean similarity	Effects belong to the full recipe.
SupCon	Positive log probabilities against the batch	Descriptive reference; no additional superiority test.

Check your understanding

Why is gradient accumulation from four batches of eight examples not equivalent to one physical batch of thirty-two? Why would a contextual win not isolate the neighborhood term? Why should loss magnitudes not be ranked across recipes? Answers: Questions 5–7.

Source trail: Contrastive and SupCon formulas; Local contextual implementation; Equation and recipe decisions; Locked v3 settings.

6 How the comparison is made fair

6.1 Specify the experiment before looking for a winner

A machine-learning result combines choices about data, representation, objective, optimization, model selection, and evaluation. If those choices are changed after seeing a preferred method's outcomes, apparent improvement can reflect the selection process rather than a reliable advantage.

The v3 protocol fixes the hypothesis, action splits, gallery, recipes, duration, tuning grid, corruption conditions, analysis, and reporting rules before selection and final-test access. This is a *prespecified* comparison. A hash-bound configuration makes the exact specification checkable; it does not replace a scientific explanation of why the choices are appropriate.

Controlled choice	What stays the same	Alternative explanation reduced
Representation	Encoder, checkpoint, preprocessing, head structure	A different backbone or input pipeline caused the difference.
Optimization exposure	Twenty epochs, physical 8×4 batches	One recipe received more updates or different within-batch information.
Initialization and order	Exact initial weights and ordered batches within seed	A favorable starting state or example order caused the difference.
Tuning budget	Three rates, three seeds per rate, same ranking rule	More search opportunities gave one method an advantage.
Evaluation input	Same gallery, queries, and per-example damage	Methods were evaluated on different problems.
Claim rule	One primary effect and fixed interval construction	A favorable metric, subset, or threshold was chosen afterward.

These controls improve **internal validity**: whether the observed difference answers the declared comparison. **External validity** asks whether it transfers to other datasets, people, actions, cameras, or damage mechanisms. Internal controls alone cannot establish that broader claim.

6.2 Pairing is more than using the same seed

Let A_s and B_s denote outcomes from two methods under a paired condition. For their difference,

$$\operatorname{Var}(A_s - B_s) = \operatorname{Var}(A_s) + \operatorname{Var}(B_s) - 2\operatorname{Cov}(A_s, B_s).$$

Shared difficulty can induce positive covariance, making the difference less variable than an unpaired comparison. Pairing is useful even when that variance reduction is not guaranteed: it prevents known input differences from becoming uncontrolled method differences.

A seed initializes a pseudorandom generator. It does not prove two execution paths consumed random values in the same order or used the same arithmetic. The project therefore stores and verifies exact starting-state and batch-plan identities. Seeds 7, 17, and 29 give three paired training realizations. They produce three separate heads per method, *not an ensemble*.

An epoch is defined as $\lceil N/32 \rceil$ balanced batches. For candidate training, $\lceil 76{,}013/32 \rceil = 2{,}376$ updates per epoch and 47,520 over twenty epochs. Class and example queues are shuffled at the epoch start and cycled within it. This is a balanced update budget, not a guarantee that every row is visited exactly once per epoch.

Gradient accumulation cannot replace the physical batch: relationships between examples in different microbatches never exist in either microbatch's similarity matrix. Adding their gradients cannot retroactively construct the missing neighbors, positives, or softmax alternatives.

6.3 Pilots, model selection, and final training are separate

Engineering pilots answer whether the implementation can complete the specified training and produce valid numerical artifacts. There is one pilot per recipe, at seed 7 and learning rate 3×10^{-4}, for twenty epochs. Checkpoints at epochs 5, 10, 15, and 20 plus epoch-zero diagnostics expose the training behavior. Their accuracy does not select a learning rate or a winning method. A pilot that happens to share a candidate's settings is still ineligible for selection.

Model selection chooses the learning rate separately for each recipe:

$$3 \text{ recipes} \times 3 \text{ rates} \times 3 \text{ seeds} = 27 \text{ fresh runs}, \qquad \eta \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}.$$

Rank candidates using unrounded mean epoch-20 clean development top-1 accuracy over all three seeds. Resolve exact ties with mean MRR, then lower learning rate. The best checkpoint is not selected across epochs. Corrupted development scores are not tuning criteria. If an endpoint learning rate wins, the grid is not expanded after the fact.

Final training starts nine fresh heads: three recipes at their selected rates, each with three paired seeds, using all 95,001 auxiliary examples. All nine must finish and be locked before novel evaluation. A final run is not a pilot continued for longer, a selected checkpoint reused unchanged, or a seed chosen because it looked favorable.

What a failed run means

A candidate needs all three valid seeds; averaging only its two successful seeds changes the comparison. Preserve failures and their causes. An allowed infrastructure retry repeats identical inputs in a new attempt; it is not a chance to draw a better seed. Numerical failure and poor accuracy are different: poor but valid accuracy is a scientific outcome, while nonfinite computation invalidates the affected evidence.

The fixed twenty-epoch duration avoids choosing a favorable stopping point. It does not claim that every method has converged or received its ideal budget. The fair statement is equal declared training and selection opportunities, with cost and limitations reported.

6.4 Why keep the final test closed?

Repeatedly looking at test results can influence human choices even if no test gradient is computed. Picking margins, a split, a corruption, or a headline after observing novel outcomes would use the test for development.

The study therefore progresses through separate authorizations: auxiliary engineering, selection, final-head locking, and only then the single novel evaluation campaign. A dataset-level opening record spans historical and current versions so a new output folder cannot silently reset the holdout's history.

After opening, scientific training and selection for this study/source remain closed. An error discovered afterward must be preserved and reported; a corrected analysis cannot silently replace the confirmatory result.

There is a historical packaging nuance. The original aggregate loader deserialized all annotations before selecting auxiliary records. The evidence supports no novel retrieval preprocessing, embedding, visualization, scoring, or tuning access; it does not support the stronger claim that no novel bytes were ever deserialized. The newer auxiliary package separates authorized inputs from novel processing. State the actual access boundary, not a cleaner-sounding history.

Source trail: Plan, Sections 4 and 9; Balanced sampler; Fixed recipes, selection, and staged authorization.

7 Designing a controlled disturbance

7.1 Where the intervention occurs

Write preprocessing as P, frozen encoding and pooling as F, and the learned head as H_m. Clean retrieval uses $H_m(F(P(x)))$. A damaged query uses

$$z_{m,c} = H_m\big(F(T_c(P(x)))\big),$$

where T_c changes the prepared pose tensor. The gallery remains clean. This location is part of the scientific question: the study perturbs pose observations, not rendered pixels, model weights, training labels, or cached embedding coordinates.

Each family starts from its own clean tensor. Damage families are not combined. After damage, there is no clipping, recentering, normalization, interpolation, or frame deletion that could alter the declared intervention.

Family	Three strengths	What changes and what stays fixed
Coordinate jitter	1%, 2.5%, 5% of torso scale	Gaussian offsets to observed x, y coordinates; confidence and missing observations unchanged.
Joint masking	3, 6, 8 joints	Zero complete trajectories, including confidence, using nested prefixes of one anatomically grouped ordering.
Frame masking	10, 25, 40 of 100 frames	Zero a contiguous nested interval across people, joints, and channels; retain all 100 time slots.

7.2 Scale jitter to the body, not arbitrary pixels

For observed joint coordinates, the intervention has the form

$$x'_{p,t,j,:2} = x_{p,t,j,:2} + \sigma\, r(x)\, \eta_{p,t,j}, \qquad \eta_{p,t,j} \sim \mathcal{N}(0, I_2).$$

σ is a fixed severity and $r(x)$ is the clean sample's lower-median positive finite torso scale. Thus the noise standard deviation is expressed relative to body size. A sample without usable torso geometry uses one fixed fallback: the lower median of valid sample-level scales from the eighty-action training set. It is measured and frozen before the relevant runs, not estimated from final-test poses or tuned to produce convenient damage.

The same Gaussian field is reused at all three severities, multiplied by the corresponding scale. That keeps the disturbance direction fixed while changing its magnitude. Missing or padded observations remain missing rather than being turned into plausible new joints.

7.3 Nested severity and shared randomness

Per-example randomness is determined by the sample identity and corruption family, not by method, training seed, or severity. Joint masks use nested prefixes; frame intervals are coupled so larger intervals contain smaller ones. Therefore, when severity increases, it strengthens the same intervention. It does not accidentally replace it with an unrelated easier random pattern.

The same damaged tensor is evaluated for every recipe and seed. A comparison then measures different responses to the same disturbance. It does not compare one model's fortunate noise draw with another model's unfortunate draw.

7.4 Validate the intervention before interpreting performance

A declared "40-frame mask" should zero the intended forty slots without shortening the sequence. Jitter should preserve confidence and scale with the declared body measure. The geometry panel checks interpretable examples and edge cases; full-query damage summaries establish what happened across the evaluation population, including fallback use and zero-damage cases.

That evidence is a manipulation check: did T_c do what the experiment says? It does not establish that contextual training handles the damage better. The October development damage audit passed its recorded execution/damage checks, but later runtime-dependent evidence still requires regeneration.

These are controlled tensor stress tests. Real occlusion may cause a pose estimator to hallucinate confident joints rather than set them to zero; real frame loss may have another temporal pattern. MotionBERT was already pretrained with masking/noise. The present experiment studies the incremental effect of head-training recipes on that shared encoder.

Check your understanding

Why exclude severity from the random seed? Why measure fallback scale only from training poses? Why is masking input coordinates not equivalent to simulating a camera occlusion? Answers: Questions 8–9.

Source trail: Corruption conventions; Corruption operators; Training-only fallback; Recorded development damage audit.

8 From retrieval ranks to a defensible conclusion

8.1 Compute ranks first, then metrics

Let r_q be the rank of the one correct gallery exemplar for query q. With a fixed gallery order to break exact cosine ties,

$$\text{Top1} = \frac{1}{Q}\sum_q \mathbf{1}[r_q = 1], \qquad \text{R@5} = \frac{1}{Q}\sum_q \mathbf{1}[r_q \le 5], \qquad \text{MRR} = \frac{1}{Q}\sum_q \frac{1}{r_q}.$$

For invented ranks $(1, 2, 7, 1)$, Top-1 is $2/4 = 0.5$, R@5 is $3/4 = 0.75$, and MRR is $(1+1/2+1/7+1)/4 \approx 0.661$. Top-1 asks whether the immediate answer is correct. R@5 asks whether the correct answer is in a short list. MRR rewards moving the correct answer closer to the top even when it is not first.

With exactly one relevant gallery item, average precision per query is $1/r_q$, so mAP equals MRR. Reporting both would not add independent evidence.

The complete final report has

$$3 \text{ recipes} \times 3 \text{ seeds} \times 10 \text{ conditions} \times 2 \text{ query definitions} = 180 \text{ cells}.$$

The ten conditions are clean plus nine corruptions. These are 180 reporting cells, not 180 independent experiments: the same heads, examples, and damage draws recur across them. The two query sets also overlap heavily.

8.2 Derive the primary effect at the query level

Let $a_{m,s,c,q} \in \{0, 1\}$ indicate a correct top-1 prediction. For each primary query, compute

$$e_q = \frac{1}{27} \sum_{s,c} \left[(a_{C,s,0,q} - a_{C,s,c,q}) - (a_{B,s,0,q} - a_{B,s,c,q}) \right],$$

where C is contextual and B is standard contrastive. Then

$$\Delta = \frac{1}{Q} \sum_{q} e_q.$$

This equals the earlier difference of mean accuracy drops because sums can be reordered. A query that becomes correct after corruption contributes a negative drop; the definition does not clip improvements to zero.

Queries have equal weight within each cell. The three seeds and nine corruption conditions have equal weight in the declared mixture. This does not imply the nine conditions are equally common in the world. It defines the specific stress mixture whose average effect is being tested.

An algebraic check connects the headline to absolute performance:

$$\Delta = (A_{C,\text{clean}} - A_{B,\text{clean}}) - (\overline{A}_{C,\text{corrupt}} - \overline{A}_{B,\text{corrupt}}).$$

Always report all three quantities. An accuracy difference of 0.03 is *three percentage points*, not a three-percent relative increase. It corresponds descriptively to thirty additional correct matches per thousand queries under the specified mixture.

8.3 Why ordinary row resampling would be misleading

The observed Δ is a number for this fixed collection of trained heads and examples. To characterize sampling variation within that setting, the protocol uses a bootstrap: repeatedly resample the observed units with replacement and recompute the statistic.

The resampling unit must reflect dependence. If several cameras recorded the same performance, resampling each view independently treats correlated evidence as if it came from separate events. The project instead samples entire performance groups, preserving all available camera views together.

The declared procedure is:

1. Within each of the twenty fixed actions, identify the performance groups.
2. Sample that action's original number of groups with replacement. A group selected twice contributes all its queries twice.
3. Use the same sampled groups across all methods, seeds, and conditions.
4. Sum their per-query effects and divide by the total sampled query count.
5. Repeat 10,000 times with the fixed generator seed, then take the 2.5th and 97.5th percentiles using the prescribed linear quantile convention.

Actions and training seeds are *not* resampled. This is not a bootstrap over all possible tasks or all possible training runs.

8.4 Unequal camera counts change the denominator

Suppose one sampled performance group contains one query with effect $+1$ and another contains three queries each with effect -1. The query-weighted mean is

$$\frac{1-1-1-1}{4} = -0.5.$$

The mean of the two group means would be $(1 + (-1))/2 = 0$, which answers a different question. Groups are resampled together to respect dependence, but queries retain their declared weighting. Resampling is not permission to change the estimand, meaning the exact quantity we intend to estimate.

Because groups can have different numbers of views, each bootstrap draw can contain a different query count. The denominator must be recomputed. Actions remain fixed strata, but their queries are not forced to have equal action weights. Notebook 3 demonstrates this with small arrays.

8.5 What the interval means and the rule for the claim

The 95% percentile interval describes uncertainty under the chosen resampling scheme. The 95% is a nominal repeated-sampling coverage target under its assumptions, not a 95% probability that the contextual recipe is superior. The single confirmatory statement is supported only if its *upper* bound is below zero. An interval spanning zero is inconclusive for that claim; it does not prove equivalence or that the true effect is exactly zero.

Observed pattern	Permitted interpretation
Upper bound below zero	Evidence of less contextual degradation under the declared mixture. Check absolute corrupted accuracy separately.
Interval includes zero	Insufficient evidence for the declared smaller-drop claim; no equivalence conclusion.
Lower bound above zero	Evidence points toward more contextual degradation under the declared mixture.

The interval's scope is conditional

It conditions on the trained heads, the three fixed seeds, fixed actions, gallery anchors, query definitions, and realized synthetic corruptions. It assumes performance groups are exchangeable within an action for this resampling purpose. Repeated performances from one person can still be dependent.

It does not quantify uncertainty over arbitrary people, new action classes, different reference clips, unrestricted training seeds, or real camera failures. Ten thousand bootstrap draws do not create ten thousand independent models.

Three seeds expose some training variation but do not justify broad population claims about random initializations. Every seed and condition is reported. SupCon, exact-official results, and per-cell comparisons remain descriptive; there are no additional unplanned superiority tests selected from favorable rows of the 180-cell table.

8.6 Sensitivity without changing the retrieval problem

Leave-one-action-out analysis asks whether one action strongly determines the reported effect. Remove that action's *queries*, retaining the original gallery and original ranks. If the gallery item were also removed, the remaining queries would face fewer alternatives, changing the retrieval problem itself. The ranges and sign changes are descriptive checks, not new confirmatory tests.

Report effect size, clean accuracy, absolute corrupted accuracy, per-seed and per-condition results, cost, and failures together. A null or adverse contextual result is successful research if the valid declared experiment is completed. A blocked experiment with no final comparison is an honest operational outcome, but it is not a null scientific result.

Check your understanding

Why resample whole performance groups but still weight individual queries? What does an interval crossing zero establish? Why do the three seeds not become 10,000 seeds through bootstrapping? Answers: Questions 10–12.

Source trail: Locked analysis, Section 6; Analysis conventions; Reproducible result/report contract.

9 Verification: what each check actually establishes

9.1 A chain of evidence, not a count of passing tests

For a final claim to be useful, the implementation must compute the intended quantity, use the intended inputs, preserve the comparison, and produce records that can be checked independently. A large software test suite supports that work, but its size is not a measure of scientific completion.

Question	Evidence that addresses it	What it does not establish
Are the equations implemented correctly?	Independent hand calculations of forward values and the declared backward rules, including edge cases	That the method works on real movements.
Does the learning path work?	A deliberately learnable synthetic problem with healthy gradients and correct self-excluded retrieval	Good generalization or real-data superiority.
Are the intended examples used?	Independently regenerated manifests, source identity, overlap/exclusion checks	Independence of every observation.
Does the adapter preserve intended encoder behavior?	Local and upstream encoder outputs on the same prepared input arrays	Byte-identical upstream preprocessing where the local confidence rule intentionally differs.
Does an execution repeat?	Fresh-process initialization, geometry, and full-cache repeats under the bound runtime	Correctness merely from repeating the same error, or equality across arbitrary hardware.
Was the intervention applied?	Fixed-panel geometry and complete-query damage summaries	Better robustness of any learned recipe.
Can results be reconstructed?	Independent projection, ranking, metric, and report replay from bound records	Permission to reinterpret the locked claim.
Can the whole study finish?	Measured end-to-end costs, retained storage, job windows, queue allowance, and researcher availability	Feasibility inferred from fast training alone.

9.2 Independent checks must be capable of disagreeing

Calling the same production function twice can establish repeatability, but both calls share its possible bug. A mathematical reference should derive a small result independently, with enough simplicity to inspect manually. For the contextual loss, both the forward formula and prescribed STE backward rule need such evidence. A second implementation copied line for line would not provide the same independence.

Likewise, a training process writing "complete" is not independent proof that the checkpoint, inputs, and scores agree. A separate validation process checks the retained outputs and their dependencies. A clean process also prevents an unrecorded in-memory state from supplying missing information.

Real pilots check finite loss, gradients, and parameters, nonzero projections, unit-vector normalization, adequate embedding variation, complete update counts, paired initialization/batches, and measured cost. The collapse check asks whether every example has become essentially the same vector; it does not certify that the remaining variation is useful. There is no real-data accuracy floor that forces the experiment to produce a favorable result.

9.3 The reproducibility problem encountered here

Floating-point operations round their results. Reordering sums, choosing a different vectorized kernel, or changing the numerical runtime can change bits even when the source expression is unchanged. A fixed random seed governs the random-number sequence; it does not fix every arithmetic path or transformation of those random numbers.

This project encountered both starting-weight reconstruction mismatches and small geometry differences across execution environments. In the geometry case, clean arrays and sample identities matched, while some derived torso scales and jitter arrays differed. Exact repeatability within each environment did not establish equality between them. Several diagnostics narrowed possibilities, but the complete historical numerical cause was not established.

The authorized response was to select and record one prospective runtime, require exact fresh-process repeatability and pairing within it, and preserve differences from older outputs. That is an explicit change in the acceptance requirement. It is not a claim that the old and new numbers were equal after all.

A later cluster kernel update failed the frozen-runtime startup check before scientific training began. Such a startup failure is evidence that the runtime binding changed, not evidence that the model produced incorrect numbers on the new kernel. The new profile was authorized for verification, but compatibility had to be measured rather than assumed.

9.4 Immutable evidence and the scope of a correction

An artifact is retained with its source, input, configuration, and runtime bindings. A corrected release writes new artifacts with explicit predecessor links instead of overwriting the failed attempt. This preserves the distinction between what originally happened and what a later correction established.

The relevant software-engineering idea is dependency-aware invalidation. A changed scientific input or implementation invalidates evidence depending on it. It need not invalidate an unrelated unchanged manifest, but old numerical outputs cannot be relabeled as new-runtime results without the required check. The protocol determines which dependencies and acceptance evidence matter.

Changes are recorded with a reason and their effect on scientific settings. The runtime amendments arose from observed reproducibility and execution problems, with earlier development outputs already in existence. They were not claimed to predate all development scores, and final novel outcomes remained unavailable. The algorithms, action IDs, corruption definitions, training duration, and claim rule stayed fixed.

9.5 Verification itself has a cost

The study discovered that nested validation repeatedly recomputed the same expensive numerical checks. Under the measured old execution path, one repeated component alone forecast over 122 hours against the old execution plan's shared 48-hour maximum-job check. That was a forecast from measured component time and call counts, not an observed 122-hour run.

The authorized optimization reuses a successful numerical verification only inside one guarded command, after rehashing its complete bound contents and matching the validator settings and identities. A changed dependency prevents reuse. Independent commands begin without inherited proof. Authorization, dataset-opening state, resources, and other dynamic checks remain fresh.

This differs from simply skipping expensive checks: the question is whether the *identical deterministic proposition* has already been established in the current command. The design still requires independent recertification and new measurements of total cost.

Storage feasibility also includes retained attempts, checkpoints, feature caches, projections, logs, temporary peaks, and a reserve. The declared rule is

$$\min(\text{filesystem free}, \text{quota remaining}) \geq 200 \cdot 2^{30} + 1.25 \cdot \text{remaining peak bytes}.$$

Free disk space outside the account's quota is not usable capacity. Nor is deleting failed scientific evidence an acceptable way to make the forecast fit. Time forecasts similarly need the complete workflow, queue/setup allowances, and researcher effort, not just the head-training loop.

Source trail: Execution-completion amendment; Validation reuse and capacity; Prospective runtime decision; Updated-kernel decision.

10 Decisions and progress through the evidence cutoff

10.1 The decisions that explain the current experiment

Decision	Reason	Consequence for the claim or process
Choose one-shot motion retrieval	Make movement comparison testable on an existing dataset with unfamiliar action labels	Evaluate matching to one clean example per novel action; do not claim a rowing-quality system.
Adopt a verified pose input contract	Available HRNet-derived annotations and documented missing captures differed from initial assumptions	Bind the actual source inventory and regenerate exact manifests.
Freeze MotionBERT and narrow to three recipes (29 Sep)	Bound the December question and resource requirements	Compare head-training recipes; preserve the broader fine-tuned v2 benchmark as unfinished historical scope.
Use primary and official query sets	Synchronized views can duplicate the gallery performance	Primary excludes all such views; official-compatible results remain descriptive.

Decision	Reason	Consequence for the claim or process
Fix duration, pairing, grid, and claim rule	Limit unequal opportunity and outcome-driven choices	Twenty epochs, three paired seeds, 27 selection runs, nine final heads, one confirmatory effect.
Omit the extra regularizer control	Keep the December study bounded before new selection outcomes	A result supports the full recipe, not the isolated neighborhood mechanism.
Complete and rehearse the full path (9 Oct)	Successful pilots did not establish that authorization through report generation worked	Successor releases and fresh affected evidence before selection.
Permit guarded verification reuse (12 Oct)	Measured repeated validation exceeded an execution limit	Preserve checks while avoiding duplicate numerical work within an identical guarded command.
Adopt one prospective runtime (19 Oct)	Historical cross-environment equality was not established	Keep exact within-runtime repeatability; record historical differences honestly.
Revalidate after kernel update (25 Oct)	The observed runtime no longer matched the accepted profile	Authorize a new profile, preserve source/settings, require fresh affected evidence.

10.2 What has actually been completed

The early work established the literature basis, prospectus, dataset/access arrangements, verified input inventory, deterministic split manifests, and adapter implementation. September extraction established encoder agreement on the same prepared arrays and two complete, matching 95,001-row feature passes. Those remain historical evidence for their recorded release.

On October 8, Week 4 closed with fresh v3 caches and all three complete twenty-epoch engineering pilots. The two full extraction passes matched exactly, and independent checks accepted the outputs. Each pilot had 47,520 updates with paired initial weights and batches. These are substantial execution milestones, but the pilot scores are ineligible for selecting a winner.

October successor work completed execution/report rehearsal and the historical full-query damage audit. The audit checked nine corruption conditions across 18,929 development queries each. Later work accepted a prospective CPU baseline, its repeatable initialization and geometry, a synthetic GPU prerequisite, and five caches under the then-frozen profile. The subsequent kernel change means those records do not automatically satisfy the new-profile requirements.

The project also has a public tracker, research journal, explanatory pose visualizations, and report/-poster drafts with results explicitly pending. Those help communicate and retain the work; they are not additional scientific evidence of recipe superiority.

10.3 What remains incomplete

At the October 28 checkpoint, Week 5 remains open and cluster jobs are held while adequate usable storage is pending. The local launcher correction and conditional remaining-work schedule are prepared. No new-profile acceptance or final comparison is inferred from that preparation.

The next scientific prerequisites are accepted new-runtime repeatability, affected caches and paired pilots, development corruption/retrieval replay, and a measured full-workload capacity decision. Only after admission can the 27 selection runs proceed, followed by nine final heads, final locks, and the novel evaluation campaign. Selection, final training, and novel opening remain closed in the latest record.

The original October 18 admission deadline was missed. November 29 final locks, December 6 computation, and December 7–18 analysis/delivery remain recorded targets; the guide does not claim the remaining schedule has been demonstrated feasible. Live capacity was not remeasured while preparing this guide.

The accurate status sentence

The study design and software have passed substantial engineering checks, including historical full pilots and damage audits. The latest runtime still needs accepted regeneration, and storage is blocking the next cluster work. There is no final evidence yet that contextual training is better, equivalent, or worse on the declared novel-action comparison.

Source trail: September execution evidence; Week 4 closure; October accepted history and kernel/storage blocker; October 28 checkpoint; Conditional remaining-work calendar.

11 Prepare to explain and defend the work

11.1 A short explanation you can adapt

About one minute

I am testing whether a particular way of training movement representations makes matching human actions less sensitive to damaged pose information. At evaluation, the system gets one clean example of each unfamiliar action and must match other clips to those examples.
I reuse a pretrained MotionBERT encoder and keep it fixed. I train the same linear comparison head with three recipes. The main comparison is the full contextual recipe against standard contrastive training. Both use paired starting weights, batches, training duration, and tuning opportunities.
I damage only the query poses, using fixed strengths of coordinate noise, missing joints, and missing frames. The primary quantity is the difference in clean-to-damaged accuracy drops, while absolute accuracy is reported too.
Historical pilots and input checks have passed. Current work is blocked on storage and new-runtime verification. The final comparison has not run, so I am not claiming that the contextual recipe wins.

Say this in your own words. The structure matters more than memorizing it: question, input/output, controlled comparison, measurement, current evidence.

11.2 A five-minute technical explanation

1. **Define the task.** A sequence of estimated 2D joint positions enters a frozen encoder, then a learned head. The output is a unit vector used for nearest-gallery retrieval. One-shot refers to one exemplar per novel action.
2. **Define the treatment.** Compare training recipes on the same head. The contextual recipe includes neighborhoods, a changed margin, and a regularizer; it is not an isolated neighborhood ablation.
3. **Explain the split and tuning.** Eighty actions train candidates; twenty other auxiliary actions select rates using clean development retrieval. Nine fresh final heads use all 100 auxiliary actions. Twenty novel actions remain outside selection and training.
4. **Explain the intervention.** Three damage families at three fixed strengths act after preprocessing and before encoding. Damage is paired across methods and nested across severities; gallery examples remain clean.
5. **Explain the statistic.** Compare clean-to-corrupt Top-1 drops, averaged over three fixed seeds and nine conditions. Resample whole performance groups within actions to preserve camera-view dependence. State the interval's conditional scope and show absolute accuracy beside the drop.
6. **Separate evidence from intent.** Describe completed historical pilots and damage checks, then the current runtime/storage prerequisite. End with what must happen before a final claim is possible.

11.3 Questions someone should be able to ask you

Why expect contextual learning to help? Relations to neighboring examples may provide useful information beyond a pair's direct similarity. That is a plausible mechanism motivating the study.

Image-domain findings do not establish robustness to noisy pose queries; this experiment tests that transfer rather than assuming it.

What exactly is original here? The backbone, pose dataset, and published objective ideas are reused. The project specifies and implements a controlled pose-retrieval comparison, its local adaptations, corruption design, protected evaluation, and verification/reporting process. It is not claiming to invent MotionBERT or contextual loss.

Why is a linear head enough? The nonlinear encoder already supplies rich motion features. A common affine head limits architectural variation while still learning a different geometry. This is a deliberate comparison setting, not proof that a nonlinear head could not perform better. Its approximately 17.8 million parameters also mean that "linear" should not be confused with negligible capacity.

Does fixed training make the comparison perfectly fair? It makes the stated budget and controls comparable. It does not guarantee equal computational cost, convergence, optimal hyperparameters, or equal compatibility with every representation. The result is conditional on these declared recipes and settings.

Why not choose the best pilot or early checkpoint? Those outputs are engineering diagnostics. Reusing them for selection or choosing an attractive epoch would give an extra, unplanned search opportunity and change the fixed-duration procedure. The declared fresh selection runs have one rule.

Why keep both official and stricter query sets? The official set enables comparison with the established benchmark convention. The primary set removes other cameras' views of the gallery performance, reducing a specific shortcut. Their distinct roles are fixed before results.

Could the model learn who the person is? Action-disjoint splits do not remove all person or setup dependence. Shared people, acquisition conditions, and representation biases remain limitations. The study is not a separate test of generalization to unseen people.

Why are three seeds not enough for a universal claim? They show three fixed training realizations and allow paired comparisons. The primary bootstrap does not resample them and cannot infer performance over all possible training randomness. Per-seed results show how consistent this small set is without disguising its limited scope.

What would count as success if contextual loses? A valid, complete, reproducible comparison with an honest contrary result. Scientific success is answering the specified question. Failing to complete the comparison because of capacity is a blocked study, not evidence of no effect.

Why has verification taken so much work? The experiment needs more than a model that trains. Source/input identity, exact pairing, damaged-input validity, runtime repeatability, and independently reconstructible results determine whether the final comparison is interpretable. The project also learned that those checks must be included in capacity planning.

What cannot you claim yet? No final recipe ranking, novel-action robustness effect, or completed empirical comparison. Successful pilots establish their execution; report drafts and a working tracker do not supply missing final results.

11.4 Draw these three things without looking

1. The shape pipeline: $2 \times 100 \times 17 \times 3$ poses to frozen features, 8,704 pooled values, a learned 2,048-dimensional unit vector, and gallery ranking.
2. The action split: eighty candidate-training actions, twenty development actions, all 100 used only after selection for fresh final training, and twenty separate novel actions for final evaluation.

3. A two-row result table containing clean accuracy, corrupted accuracy, and drop. Construct a case where contextual degrades less but has worse corrupted accuracy, then explain why both facts belong in the report.

11.5 Use the notebooks to test understanding

Lab	Concept	What you should explain afterward
01	Cosine geometry and a normalized-head gradient	Why direction determines ranking; how changing W changes it; what finite differences verify.
02	Splits, leakage, pairing, and equal selection budgets	Why repeated views and unequal tuning opportunities can distort a comparison.
03	Degradation and paired cluster bootstrap	Why smaller drop can coexist with worse accuracy; why the sampled query count is the denominator.

All examples are invented. Do not report their scores as project results. Notebook 1 simplifies the training objective to make differentiation visible; it does not implement or validate the contextual STE. Notebook 3 implements the analysis logic on a tiny synthetic population, not the locked real-data campaign.

A Answers to the study checks

1. **Random row splits.** They can put the same action classes in training and evaluation, testing new clips of familiar actions rather than transfer to unfamiliar actions. They can also separate synchronized views of one performance across roles.
2. **Gallery exclusion.** A different camera file can still depict the same event as the gallery example. The primary set excludes the underlying performance group, not only one filename.
3. **Learning with a frozen encoder.** The encoder's output stays fixed, but the head's W, b change how those features map into comparison directions. A novel action does not need a learned output-class coordinate: it supplies a gallery exemplar and is recognized by similarity to it.
4. **Cache perturbation.** A pose corruption is transformed nonlinearly by the encoder. In general $F(T(X))$ is not equivalent to adding noise to $F(X)$. The intervention location is therefore part of the hypothesis.
5. **Physical batch.** Pairwise similarities, contextual neighbors, and SupCon denominators depend on the examples simultaneously present. Summing gradients from separate smaller batches omits cross-microbatch relations.
6. **Full recipe.** Contextual and standard contrastive differ in the neighborhood term, positive margin, and similarity regularizer. A matched ablation would be needed to isolate one component's causal contribution.
7. **Loss magnitudes.** The formulas, reductions, coefficients, and attainable minima differ. Compare retrieval under the common evaluation, not which method reports the smallest numerical training loss.
8. **Shared severity randomness.** Reuse of the underlying noise field or nested mask makes a stronger level a controlled strengthening of the same damage. Unrelated draws could obscure the effect of increasing severity.
9. **Fallback and realism.** A training-only fallback avoids estimating preprocessing parameters from held-out poses. Synthetic zero masks and Gaussian noise do not reproduce every output a real pose estimator produces under camera occlusion, so they support a narrower robustness claim.
10. **Groups versus query weights.** Groups preserve dependence during resampling; query weights preserve the chosen statistic. Equal weighting of unequal-sized group means would change the quantity being estimated.
11. **Interval crossing zero.** The data do not support the prespecified smaller-drop claim at that rule. This is neither proof of equality nor an estimate that the probability of no effect is 95%.
12. **Bootstrap draws and seeds.** Repeatedly resampling performance groups recombines evidence from the same three heads per method. It creates no new independent training realizations or uncertainty over unobserved seeds.

B Notation and essential terms

Symbol or term	Meaning here
X	Prepared pose tensor for one capture.
$h \in \mathbb{R}^{8704}$	Frozen, pooled, unnormalized pre-head feature vector.
W, b	Trainable affine head weights and bias.
$u \in \mathbb{R}^{2048}$; z	Head output before normalization; its unit-length direction.
s_{ij}; d_{ij}	Cosine similarity; squared unit-vector distance.
y_{ij}	Indicator that two training examples have the same action label.

Symbol or term	Meaning here
P, K, n	Classes per batch, examples per class, and physical batch size.
k, ϵ, α	Contextual neighbor rank, membership tolerance, and surrogate-gradient factor.
τ; η	SupCon temperature; optimizer learning rate.
$A_{m,s,c}$; $d_{m,s,c}$	Top-1 accuracy and clean-to-corrupt drop; this d has method/seed/condition indices, unlike pairwise distance d_{ij}.
Δ	Contextual drop minus contrastive drop, averaged over fixed seeds and corruptions. Negative favors less contextual degradation.
Gallery; query	Labeled reference examples; the example to be matched.
Embedding	A learned numerical representation used for comparison.
Epoch	The prescribed number of balanced updates, not necessarily one visit to each row.
Hyperparameter	A setting chosen outside gradient updates, such as learning rate; distinct from learned weights.
Leakage	Information from an evaluation role influences development or makes the evaluation easier in a way the claimed design excludes.
Estimand	The exact quantity the experiment intends to estimate.
Bootstrap	Recompute a statistic over repeated resamples of the declared units.
Provenance	The source, input, configuration, and runtime identities connecting an output to the execution that produced it.
Gate	An explicit requirement and its acceptance evidence before later work is eligible.

C Source map and evidence boundaries

The linked repository sources are pinned to commit `b32d50f66206b561fa34867c9274fac6ba7c287d`, the main-branch record through October 25, unless marked as the October 28 preparation record at `5d50410b07e19a980fe3b711e63b10417b40b818`. This keeps the account traceable even if a live branch later changes. The guide is explanatory documentation; the canonical protocol and authorized amendments control execution.

Read this for	Primary project source
Scientific question and rules	Final v3 plan and protocol v3.
Machine-readable settings	Protocol template and 27 candidate configurations. A template is not an executed or resolved design lock.
Method equations and adaptation	Contextual source notes, contextual loss, and pairwise losses.
Reuse and preprocessing	Upstream manifest, notices, and adapter acceptance.
Why the scope changed	Decision 0004; 0006, 0007, 0008, and 0009.
Accepted history and current state	Week 4, Week 5 through October 25, and October 28 current checkpoint.

The October 28 schedule's opening prose overstates the October 25 filesystem failure. The numerical Week 5 record shows quota below the requirement while filesystem free space exceeded it; usable space was still insufficient because the minimum governs. This guide uses the numerical record and states the storage blocker without treating an old observation as live capacity.

The intended prerequisites are general calculus, linear algebra, elementary probability/statistics, and strong programming. No transcript, grades, student identifier, or private academic record is included. Figures are schematic and notebook data are synthetic. No licensed pose array, learned checkpoint, participant record, or private operational log is included.

References

[1] J. Liu et al. *NTU RGB+D 120: A Large-Scale Benchmark for 3D Human Activity Understanding.* IEEE TPAMI, 2020. Authors' paper; Official dataset. The local input contract separately identifies the adopted HRNet-derived 2D annotations; the dataset's title does not make our input coordinates 3D.

[2] W. Zhu et al. *MotionBERT: A Unified Perspective on Learning Human Motion Representations.* ICCV, 2023. Official paper and supplement. Read for the encoder's original purpose; our frozen-head comparison is a different experiment from its downstream fine-tuning results.

[3] C. Liao, T. Tsiligkaridis, and B. Kulis. *Supervised Metric Learning to Rank for Retrieval via Contextual Similarity Optimization.* ICML, 2023. Paper and appendix. Read for the contextual objective's origin. The local source notes resolve equation/backward conventions and document pose-study adaptations.

[4] P. Khosla et al. *Supervised Contrastive Learning.* NeurIPS, 2020. Official paper. Read for label-based contrastive learning; this study fixes a single-view variant.